$0 \le tr(\rho W)$ $\quad W = \frac{1}{2} I - |B \rangle\langle B|$

Entanglement detection using Bell (correlation) measurement

$$|B_1\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) \qquad |B_1\rangle\langle B_1| = \frac{1}{4}\left(II + XX - YY + ZZ\right)$$

$|\langle 00|B_1\rangle|^2 = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$

$$|B_2\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right) \qquad |B_2\rangle\langle B_2| = \frac{1}{4}\left(II + XX + YY - ZZ\right)$$

$$|B_3\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle - |11\rangle\right) \qquad |B_3\rangle\langle B_3| = \frac{1}{4}\left(II - XX + YY + ZZ\right)$$

$$|B_4\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right) \qquad |B_4\rangle\langle B_4| = \frac{1}{4}\left(II - XX - YY - ZZ\right)$$

$p_i = tr\left(\rho\, |B_i\rangle\langle B_i|\right)$ $\quad i = 1,2,3,4$

$$\boxed{0 \le p_i \le 1, \quad \sum_{i=1}^{4} p_i = 1}$$

Prob-space (Tetrahadron)

$p_4 = 1 - (p_1 + p_2 + p_3)$

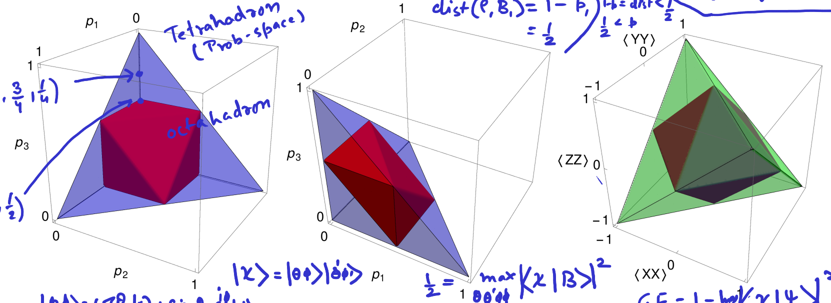
- If we get a p_i such that $\frac{1}{2} < p_i \le 1$ then we say ρ is Entangled
- If we get all p_i such that $\boxed{0 \le p_i \le \frac{1}{2}}$ then we cannot say anything about Ent. of ρ

octahadron

$dist(\rho, B_1) = 1 - p_1$ $= \frac{1}{2}$

$1 - p = dist < \frac{1}{2}$

$\frac{1}{2} < p$



Tetrahadron (Prob-space)

octahadron

ρ

$\vec{p} = (0, 0, \frac{3}{4}, \frac{1}{4})$

ρ'

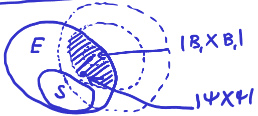
$\vec{p}' = (0, 0, \frac{1}{2}, \frac{1}{2})$

$|\chi\rangle = |\theta\phi\rangle|\theta'\phi'\rangle$

$|\theta\phi\rangle = \cos\theta/_2 |0\rangle + \sin\theta/_2\, e^{i\phi} |1\rangle$

$\frac{1}{2} = \max_{\theta\theta'\phi\phi'} |\langle \chi | B \rangle|^2$

$GE = 1 - \max_{\theta\theta'\phi\phi'} |\langle \chi | \psi \rangle|^2$



video

Dr. Arun Sehrawat

$CNOT^{-1} = CNOT$

$XZ = -iY$

Universal set of "discrete" gates for multi-qubits

$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

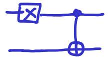
Bit flip Error

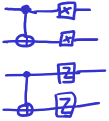
$4^2 = 16$

- Generators of the multi-qubit Clifford group: $\{H, S, \text{CNOT}\}$

$CNOT\,(X \otimes I) = (X \otimes X)\, CNOT$

Control





$\text{CNOT}\,(X \otimes I)\,\text{CNOT} = X \otimes X$	$\text{CNOT}\,(I \otimes X)\,\text{CNOT} = I \otimes X$
$\text{CNOT}\,(Y \otimes I)\,\text{CNOT} = Y \otimes X$	$\text{CNOT}\,(I \otimes Y)\,\text{CNOT} = Z \otimes Y$
$\text{CNOT}\,(Z \otimes I)\,\text{CNOT} = Z \otimes I$	$\text{CNOT}\,(I \otimes Z)\,\text{CNOT} = Z \otimes Z$

Phase flip

Rot in $\mathbb{R}^3$

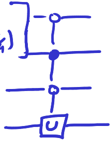
$CNOT\,(I \otimes Z) = Z \otimes Z\,(CNOT)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$[\,|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X\,] \qquad \underbrace{X \otimes I\,[\,|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X\,]}_{[\,|1\rangle\langle 0| \otimes I \;+\; |0\rangle\langle 1| \otimes X\,]}$$

$$T = \begin{bmatrix} 1 & 0 \\ 0 & \exp(i\pi/4) \end{bmatrix}$$

$\pi/8$ gate



$$\begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= |1\rangle\underbrace{\langle 1|1\rangle}_{1}\langle 0| \otimes X + |0\rangle\underbrace{\langle 0|0\rangle}_{1}\langle 1| \otimes X = \underbrace{[\,|1\rangle\langle 0| + |0\rangle\langle 1|\,]}_{X} \otimes X \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

- Generators of the multi-qubit Unitary group: $\{H, T, \text{CNOT}\}$ up to a desired accuracy

- An arbitrary unitary operator can be expressed **exactly** as a product of (two-level) unitary operators, where each acts non-trivially only on a **two-dimensional** subspace.

- There exist unitary transforms which require exponentially many gates to approximate.

M. A. Nielsen and I. L. Chuang, *Quantum Compu and Quantum Information*, Chapt 4.